

10(a)

any **nine** from:

gene mutation

- 1 base substitution ;
- 2 (often) does not have a significant effect on phenotype / silent mutation ;
- 3 base, insertion / deletion leads to, frame shift / described ;
- 4 (so) has significant effect on phenotype ;
- 5 change in, primary structure / amino acid sequence / polypeptide made ;
- 6 change in, tertiary structure / 3D shape / folding ;
- 7 loss of function in protein or enzyme / example described ;
- 8 (premature) stop codon ;

Huntinton's disease

- 9 (mutant allele) is dominant ;
- 10 HD / dominant, allele has more repeats of base triplet CAG (than normal) ;
- 11 heterozygote will have disease ;
- 12 brain cells die more rapidly (than normal) / brain degeneration ;
- 13 involuntary movements / mental deterioration or described / mood changes ;
- 14 onset in middle age / *idea that* no change in phenotype in earlier life ;
- 15 AVP ; e.g. greater number of CAG repeats affects, earlier onset / severity of disease

Question	Answer	Marks
10(b)	<i>any six from:</i> 1 DELLA proteins inhibit, transcription factor / PIF; 2 gibberellin binds to receptor ; 3 in aleurone layer ; 4 <i>ref. to</i> enzyme involved ; 5 DELLA proteins broken down ; 6 TF / PIF, binds to promoter region (of DNA) ; 7 transcription of gene coding for amylase / AW ; 8 <i>ref. to</i> translation (leading to amylase production) ;	6